JAVA PRACTICAL PROGRAM REPORT

Interface Inheritance and Multiple Interfaces

Four Java programs combined into one report

Language	Java
Programs	4
Topics	Interface inheritance, implementing multiple interfaces, method implementation, object creation
File names	InterfaceInheritance.java MultipleInheritance.java MultipleInterface.java MultipleInterfaceInheritance.java

General note: Each program contains its own public class. Save and compile each program in a separate file using the exact filename shown above.

1. Aim

To study and demonstrate Java interfaces, interface inheritance, and how a class can implement one or more interfaces.

2. Requirements

• Java Development Kit (JDK) with javac and java commands

• A text editor or Java IDE

• One .java file per public class

3. Key Concepts

Concept	Meaning
Interface	A contract that declares methods a implementing class must provide (for these examples, methods are implicitly public and abstract).
extends	Used by an interface to inherit from another interface; an interface can extend more than one interface.
implements	Used by a class to implement an interface and provide the required method bodies.
Multiple interfaces	A class can implement several interfaces, combining their method requirements.
public methods	Implemented interface methods are declared public because interface methods are public.

Program 1 — Interface Inheritance

Aim: To demonstrate one interface extending another interface, and a class implementing the child interface.

Concept: An interface can extend another interface. The BMW class implements Car, so it must provide implementations for the methods declared by both Car and its parent Vehicle.

Algorithm

1. Declare Vehicle with the start() method.
2. Declare Car extending Vehicle and add drive().
3. Create BMW implementing Car and implement start() and drive().
4. In main(), create a BMW object and call both methods.

Program Flow

main() → create BMW object → start() → drive() → end

Source Code

```
interface Vehicle {
    void start();
}

interface Car extends Vehicle {
    void drive();
}

class BMW implements Car {

    public void start() {
        System.out.println("Car starts");
    }

    public void drive() {
        System.out.println("Car is driving");
    }
}

public class InterfaceInheritance {
    public static void main(String[] args) {

        BMW b = new BMW();

        b.start();
        b.drive();
    }
}
```

Code Explanation

Code element	Explanation
interface Vehicle	Declares the start() method.
interface Car extends Vehicle	Car inherits the start() contract and declares drive().
class BMW implements Car	BMW agrees to implement the methods required by Car and Vehicle.
public void start() / drive()	These methods provide the actual behavior and print messages.
BMW b = new BMW()	Creates a BMW object.

Code element	Explanation
b.start(); b.drive();	Calls the two implemented methods in order.

Compilation and Execution

```
javac InterfaceInheritance.java
java InterfaceInheritance
```

Console Output

```
Car starts
Car is driving
```

Execution Status: PASS

Program 2 — Multiple Inheritance Through Interfaces

Aim: To demonstrate a class implementing two interfaces.

Concept: Java classes cannot extend multiple classes, but a class can implement multiple interfaces. Document provides implementations for both Printable and Showable.

Algorithm

1. Declare Printable with print().
2. Declare Showable with show().
3. Create Document implementing both interfaces.
4. Implement print() and show().
5. In main(), create a Document object and call both methods.

Program Flow

> main() → create Document object → print() → show() → end

Source Code

```
interface Printable {
    void print();
}

interface Showable {
    void show();
}

class Document implements Printable, Showable {

    public void print() {
        System.out.println("Printing document");
    }

    public void show() {
        System.out.println("Showing document");
    }
}

public class MultipleInheritance {
    public static void main(String[] args) {

        Document d = new Document();

        d.print();
        d.show();
    }
}
```

Code Explanation

Code element	Explanation
interface Printable / Showable	Declare separate method requirements.
implements Printable, Showable	The class implements both interfaces, separated by a comma.
print() and show()	Provide the behavior required by each interface.
Document d = new Document()	Creates a Document object.

Code element	Explanation
d.print(); d.show();	Invokes both methods in order.

Compilation and Execution

```
javac MultipleInheritance.java
java MultipleInheritance
```

Console Output

```
Printing document
Showing document
```

Execution Status: PASS

Program 3 — Multiple Interfaces

Aim: To demonstrate a Mobile class implementing Camera and Music interfaces.

Concept: A single class can implement multiple interfaces, combining different capabilities in one object.

Algorithm

1. Declare Camera with takePhoto().
2. Declare Music with playMusic().
3. Create Mobile implementing both interfaces.
4. Implement both methods.
5. In main(), create a Mobile object and call both methods.

Program Flow

main() → create Mobile object → takePhoto() → playMusic() → end

Source Code

```
interface Camera {
    void takePhoto();
}

interface Music {
    void playMusic();
}

class Mobile implements Camera, Music {

    public void takePhoto() {
        System.out.println("Mobile takes photo");
    }

    public void playMusic() {
        System.out.println("Mobile plays music");
    }
}

public class MultipleInterface {
    public static void main(String[] args) {

        Mobile m = new Mobile();

        m.takePhoto();
        m.playMusic();
    }
}
```

Code Explanation

Code element	Explanation
interface Camera / Music	Declare photo-taking and music-playing capabilities.
class Mobile implements Camera, Music	Mobile implements both sets of requirements.
takePhoto() / playMusic()	Print messages representing the phone's two capabilities.
Mobile m = new Mobile()	Creates a Mobile object.

Code element	Explanation
m.takePhoto(); m.playMusic();	Calls each capability in order.

Compilation and Execution

```
javac MultipleInterface.java
java MultipleInterface
```

Console Output

```
Mobile takes photo
Mobile plays music
```

Execution Status: PASS

Program 4 — Multiple Interface Inheritance

Aim: To demonstrate an interface extending two interfaces, and a class implementing the combined interface.

Concept: Interface C extends both A and B, so it inherits methodA() and methodB() and also declares methodC(). Demo implements C and therefore implements all three methods.

Algorithm

1. Declare interface A with methodA().
2. Declare interface B with methodB().
3. Declare interface C extending A and B, and add methodC().
4. Create Demo implementing C and implement all three methods.
5. In main(), create a Demo object and call all methods.

Program Flow

main() → create Demo object → methodA() → methodB() → methodC() → end

Source Code

```
interface A {
    void methodA();
}

interface B {
    void methodB();
}

interface C extends A, B {
    void methodC();
}

class Demo implements C {

    public void methodA() {
        System.out.println("Method A");
    }

    public void methodB() {
        System.out.println("Method B");
    }

    public void methodC() {
        System.out.println("Method C");
    }
}

public class MultipleInterfaceInheritance {
    public static void main(String[] args) {

        Demo d = new Demo();

        d.methodA();
        d.methodB();
        d.methodC();
    }
}
```

Code Explanation

Code element	Explanation
interface A / B	Each declares one method.
interface C extends A, B	C inherits the method requirements of both parent interfaces and declares methodC().
class Demo implements C	Demo must implement methodA(), methodB(), and methodC().
Demo d = new Demo()	Creates an object of Demo.
d.methodA(); d.methodB(); d.methodC();	Calls all three methods in order.

Compilation and Execution

```
javac MultipleInterfaceInheritance.java
java MultipleInterfaceInheritance
```

Console Output

```
Method A
Method B
Method C
```

Execution Status: PASS

8. Comparison of the Four Programs

Program	Inheritance / Implementation	Methods called
1. InterfaceInheritance	Car extends Vehicle; BMW implements Car	start(), drive()
2. MultipleInheritance	Document implements Printable and Showable	print(), show()
3. MultipleInterface	Mobile implements Camera and Music	takePhoto(), playMusic()
4. MultipleInterfaceInheritance	C extends A and B; Demo implements C	methodA(), methodB(), methodC()

9. Result and Conclusion

All four programs demonstrate Java interface behavior. The first shows an interface extending another interface; the second and third show a class implementing two interfaces; and the fourth shows one interface extending two parent interfaces. The output confirms that the classes implement the required methods and that method calls execute in the order written in main().

Note: Each source code is a separate Java program because each defines a public class. Do not paste all four public classes into one .java file.